

Samarth Shete

AI Engineer

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SUMMARY

AI Engineer and Software Development Engineer with 3+ years of experience designing scalable AI platforms, intelligent backend systems, and cloud-native applications across enterprise environments. Specialized in Generative AI, Retrieval-Augmented Generation (RAG), LLM evaluation, AI agent frameworks, Python, FastAPI, AWS, and modern microservices architecture, delivering secure, production-grade AI solutions.

SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, SQL, Bash

AI Engineering & LLM Systems: Large Language Models (LLMs), Generative AI, Retrieval-Augmented Generation (RAG), AI Agents, Agentic Workflows, Prompt Engineering, Prompt Security, LLM Evaluation, LLM-as-a-Judge, AI Guardrails, LLM Observability

Machine Learning & NLP: PyTorch, Scikit-learn, Hugging Face Transformers, Natural Language Processing (NLP), Embeddings, Semantic Search, Vector Similarity Search, Feature Engineering, Model Evaluation, Fine-Tuning Fundamentals

Software Engineering: Object-Oriented Programming (OOP), Data Structures & Algorithms, Design Patterns, Backend Development, RESTful APIs, Microservices Architecture, Asynchronous Programming, API Integration, Secure API Development

Frameworks & Libraries: FastAPI, Node.js, Express.js, React, LangChain, OpenAI API, Pydantic

Data Engineering & Databases: PostgreSQL, MongoDB, Neo4j, Redis, pgvector, Vector Databases, Data Modeling

Cloud & DevOps: AWS (EC2, Lambda, EKS, RDS, S3), Docker, Kubernetes, GitHub Actions, CI/CD Pipelines

MLOps & Observability: ML Pipeline Development, Model Monitoring, OpenTelemetry, Prometheus, Grafana, Structured Logging

Testing & Security: AI Red Teaming, Prompt Injection Detection, Adversarial Testing, Application Security, Authentication & Authorization (JWT), Automated Testing, Performance Testing, Regression Testing

EXPERIENCE

Adobe, VA | AI Engineer

Mar 2026 – Present

- Architected a production-grade RAG pipeline using LangChain, OpenAI API, and pgvector on AWS RDS, processing 50K+ enterprise documents with semantic search, which improved information retrieval accuracy by 42%.
- Built an autonomous AI agent framework with FastAPI and Python, implementing agentic workflows for document analysis and summarization, handling 10K+ daily requests, which automated 70%.
- Deployed LLM guardrails and prompt injection detection using custom adversarial testing frameworks, reducing security vulnerabilities by 65% and achieving 99.2% detection rate against malicious prompts across production models.
- Designed and implemented an LLM-as-a-Judge evaluation pipeline with structured logging via OpenTelemetry and Prometheus, monitoring 15+ performance metrics across 3 model versions, which accelerated model iteration cycles by 55%.

Virtual Infotech Solution, India | Software Development Engineer

May 2023 – Jul 2024

- Designed predictive machine learning models using Scikit-learn and XGBoost for financial risk analysis, improving fraud detection precision by 14.8% while maintaining strict latency requirements.
- Architected a microservices-based document processing platform on AWS (EC2, Lambda, S3, RDS) using FastAPI and Python, processing 150K+ documents monthly, which reduced processing latency by 58%.
- Built an automated CI/CD pipeline using GitHub Actions and Docker, deploying containerized microservices to AWS EKS with zero-downtime rolling updates, reducing deployment time by 75% and increasing release frequency from bi-weekly to daily.
- Developed a real-time model monitoring dashboard with Prometheus and Grafana, tracking 20+ operational metrics and configuring intelligent alerts, which reduced mean-time-to-detection (MTTD) by 60%.
- Implemented asynchronous processing workflows using Python asyncio and Redis caching, optimizing API response times for LLM-powered features, achieving 3.2x throughput improvement and reducing average response latency from 1.2s to 380ms.

Neon IT Systems, India | Software Engineer

Jun 2021 – Apr 2023

- Engineered a full-stack web application with React, Node.js, and PostgreSQL, serving 30K+ monthly active users, which increased user engagement by 35% and reduced page load time by 50% through optimized bundling.
- Containerized the entire application stack using Docker and managed container orchestration with Kubernetes, improving development environment parity and reducing production deployment failures by 55%.
- Built an automated regression testing suite with Jest and Supertest, integrated into the CI/CD pipeline, achieving 87% test coverage and reducing post-deployment bug rates by 48%.
- Optimized SQL queries and implemented caching strategies with Redis, reducing database load by 40% and improving average database query response time from 850ms to 180ms.
- Led code review and established software engineering best practices (OOP, design patterns, SOLID principles) across a 5-member team, resulting in 30% reduction in technical debt and improving code maintainability scores by 25%.

EDUCATION

Master of Science in Computer Science

May 2026

George Washington University, USA

Bachelor of Technology in Computer Engineering

May 2021

Sanjay Bhokare Group of Institutes, India